

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

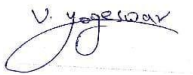
Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I V Yogeswar(192425175) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

V Yogeswar
Name of the student:

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

Answer:

A high initial learning rate helps the model move quickly toward the minimum loss, reducing training time. As the learning rate is gradually decreased (learning rate decay), the model takes smaller steps, allowing it to converge more accurately.

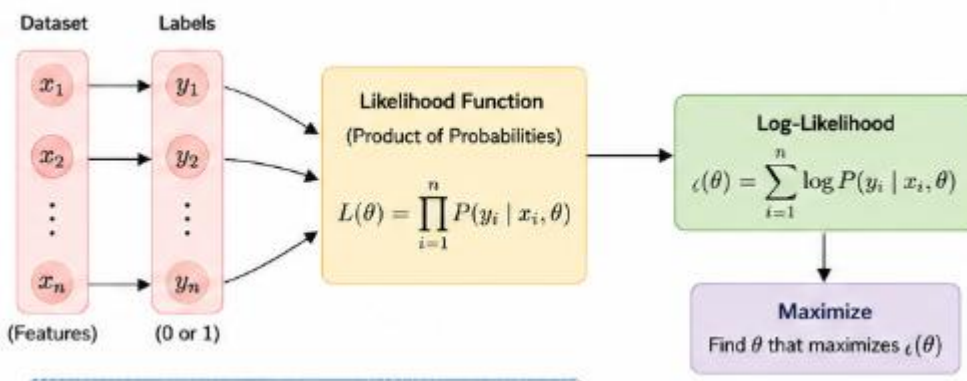
This strategy performs better than a constant learning rate when:

- The loss function is smooth or convex.
- The initial learning rate is not excessively high.
- Learning rate decay is applied properly.

Risks:

- A very high learning rate may overshoot the minimum.
- It may cause unstable training or divergence.
- Improper decay can slow convergence or prevent reaching the optimum.

Conclusion: A high initial learning rate with gradual decay provides faster convergence and better optimization than a constant learning rate when chosen appropriately.



2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers

- The sample size increases
What does this imply about the robustness of MLE?

Answer:

For a Gaussian distribution with known variance, the Maximum Likelihood Estimator (MLE) of the mean is:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

where n is the number of samples.

Effect of Outliers

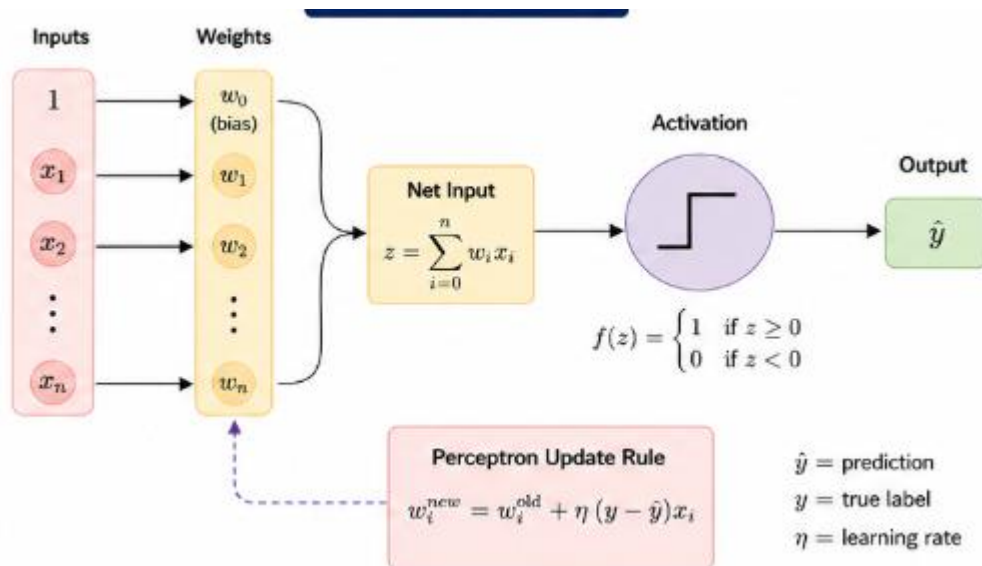
- Outliers significantly change the sample mean.
- Therefore, the MLE estimate shifts toward the outliers.
- This shows that MLE is sensitive to extreme values.

Effect of Increasing Sample Size

- As the sample size increases, the estimate becomes more accurate.
- The variance of the estimator decreases.
- The estimate approaches the true mean.

Conclusion

MLE is efficient for large datasets but is not robust to outliers because the mean is highly affected by extreme observations.



3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

Answer:

Challenges

- Very high dimensionality (10,000 features).
- Small number of training samples (500).
- High risk of overfitting.
- Increased computational complexity.

Proposed Pipeline

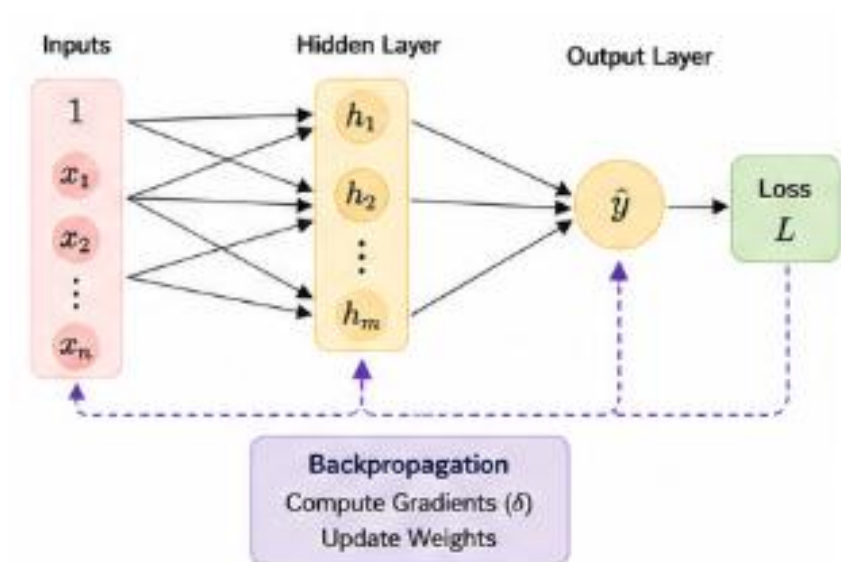
1. Data preprocessing (normalization).
2. Feature selection to remove irrelevant features.
3. Dimensionality reduction using PCA.
4. Train a regularized classifier (L1/L2 regularization).
5. Validate using cross-validation.

Justification

- Feature selection reduces noise.
- PCA reduces feature dimensions while preserving information.
- Regularization limits model complexity.
- Cross-validation improves model reliability.

Conclusion

This pipeline reduces overfitting and improves generalization.



4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

Answer:

Adding more hidden neurons increases the network's ability to learn complex nonlinear relationships because more neurons can represent more features and decision boundaries.

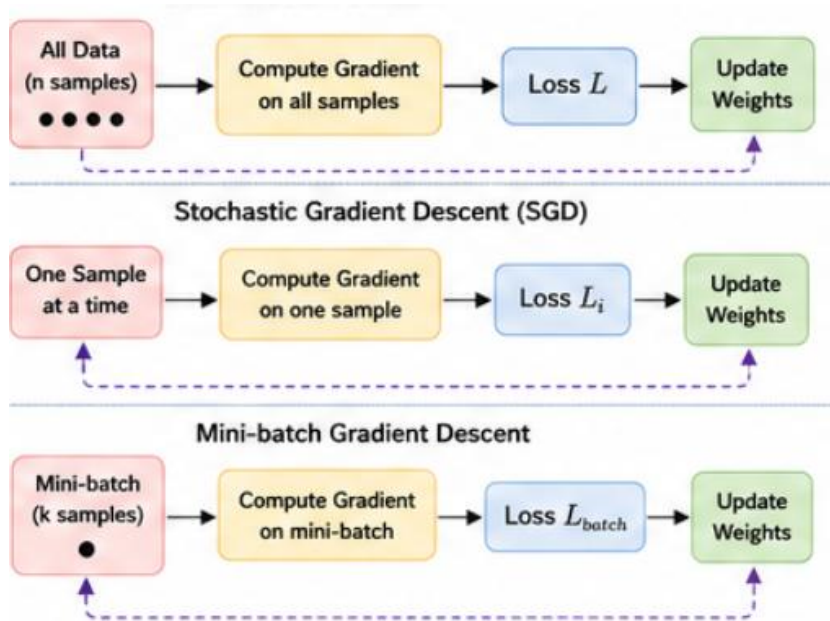
However, increasing neurons does not always improve performance because:

- The model may memorize the training data (overfitting).

- Generalization to unseen data decreases.
- Training time and computational cost increase.

Conclusion

An optimal number of hidden neurons provides a balance between representational power and generalization.



5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

Answer:

Impact on Gradient Updates

- High-dimensional data produces noisy and unstable gradients.
- Backpropagation becomes less efficient.

Impact on SGD Convergence

- SGD requires more iterations to converge.
- Optimization becomes slower due to many parameters.

Impact on Generalization

- The model is more likely to overfit.
- Prediction accuracy on unseen data decreases.

Techniques to Overcome These Issues

1. Dimensionality Reduction (PCA)

- Removes redundant features.
- Reduces computation and improves convergence.

2. Regularization (L1/L2 or Dropout)

- Prevents overfitting.
- Improves model generalization.

Additional Techniques

- Feature selection.
- Batch Normalization.
- Early stopping.

Conclusion

Reducing dimensionality and applying regularization improve convergence speed, stabilize training, and enhance generalization in high-dimensional datasets.

